

ColePago Student Wallet

ColePago Wallet AI Features & Technology Stack

AI & Smart Features

- **Face Detection (Google ML Kit):**
 - Uses the selfie camera to detect faces.
 - Ensures only one face is present for secure transactions.
 - Obscures actions if multiple faces are detected.
- **Biometric Authentication:**
 - Supports fingerprint, face, and iris recognition.
 - PIN fallback if biometrics unavailable.
- **Notifications:**
 - Real-time transaction and activity alerts for parents and kids.

Payment & Security

- **NO REAL MONEY EXPOSURE IN BACKEND (CRUCIAL SECURITY):**
- **ONLY VIRTUAL TOKENS/COINS FLOW THROUGH THE BACKEND.**
 - All real money transactions (card payments, wallet funding) are securely handled by Stripe & Mercado Pago.
 - The backend never stores or processes real money or card data just virtual wallet balances and token transactions.
- **Stripe & Mercado Pago Integration:**
 - Secure, modern payment SDKs for wallet funding and transactions.
- **Parent/Child Authentication:**
 - Role-based access and secure login for both parents and children.

Backend Stack

- **FastAPI:** High-performance Python backend for APIs.
- **PostgreSQL:** Reliable relational database for wallet and transaction data.

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- ****Docker:**** Containerized deployment for easy scaling and portability.
- ****Terraform:**** Infrastructure as code for cloud provisioning and management.

Mobile App Stack

- ****Flutter:**** Cross-platform mobile app framework (Android/iOS).
- ****Dart:**** Modern language powering the Flutter app.
- ****Camera & ML Kit:**** For face detection and camera-based features.
- ****Local Auth:**** For biometric authentication.
- ****Shared Preferences:**** For secure local storage.
- ****Dio & HTTP:**** For robust API communication.

Visual Design & Promotion

- Colorful, kid-friendly branding with backpack logo
- Modern UI/UX for both parents and kids
- Secure, trustworthy, and easy to use

For more info, contact us at: [your-email@example.com]

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ColePago/StudentWallet Secure Digital Wallet for Schools

![ColePago Logo](https://dummyimage.com/120x60/4e73df/ffffff&text=ColePago)

How It Works

1. Parent Funds Wallet

![Parent Funding](https://dummyimage.com/100x80/6c757d/ffffff&text=Parent+Funds)

- Parent uses the app to add money via Stripe/Mercado Pago.

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- ****No card data or real money ever touches our backend.****

2. Virtual Tokens Credited

![Tokens](https://dummyimage.com/80x80/36b37e/ffffff&text=Tokens)

- Backend receives payment confirmation and credits virtual tokens/coins to the child's wallet.
- Only tokens/coins flow through the backend.

3. Child Spends at School

![Child Pays](https://dummyimage.com/100x80/f6c23e/ffffff&text=Child+Pays)

- Child uses tokens to pay at school canteen, store, or for activities.
- Parent receives instant notification.

4. Security & Protection

![Security](https://dummyimage.com/80x80/1cc88a/ffffff&text=Secure)

- ****AI Face Detection & Biometrics:**** Only the right child can access the wallet.
- ****No Real Money Exposure:**** All real money handled by PCI-compliant providers.
- ****Data Privacy:**** No sensitive payment data stored in the app or backend.

****ColePago/StudentWallet**** Safe, smart, and simple for families and schools.

Contact: [your-email@example.com]

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ColePago/StudentWallet Secure Architecture Diagram

![ColePago Logo](https://dummyimage.com/120x60/4e73df/ffffff&text=ColePago)

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****Key Security Highlight:****

- The backend never handles real money only virtual tokens/coins.
- All real money is processed by Stripe/Mercado Pago (PCI-compliant providers).
- This ensures maximum safety for schools, parents, and kids.